

Subject Description Form

Subject Code	EIE4435
Subject Title	Image and Audio Processing
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	EIE3312 Linear Systems or EIE3103 Digital Signals and Systems
Objectives	To provide a broad treatment of the fundamentals in image and audio processing.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Understand the fundamentals of image and audio signal processing and associated techniques. 2. Understand how to solve practical problems with some basic image and audio signal processing techniques. 3. Have the ability to design simple systems for realizing some multimedia applications with some basic image and audio signal processing techniques. <u>Category B: Attributes for all-roundedness</u> 4. Present ideas and findings effectively. 5. Learn independently.
Subject Synopsis/ Indicative Syllabus	Syllabus: 1. <u>Image processing</u> 1.1 Fundamentals of digital image: Digital image representation and visual perception, image sampling and quantization. 1.2 Image enhancement: Histogram processing; Median filtering; Low-pass filtering; High-pass filtering; Spatial filtering; Linear interpolation, Zooming. 1.3 Image coding and compression techniques: Scalar and vector quantizations; Codeword assignment; Entropy coding; Transform image coding; Wavelet coding; Codec examples. 1.4 Image analysis and segmentation: Feature extraction; Histogram; Edge detection; Thresholding. 1.5 Image representation and description: Boundary descriptor; Chaincode; Fourier descriptor; Skeletonizing; Texture descriptor; Moments. 2. <u>Audio processing</u> 2.1 Fundamentals of digital audio: Sampling; Dithering; Quantization; psychoacoustic model; A/D and D/A conversions; Equalisation. 2.2 Audio compression: Critical bands; Threshold of hearing; Amplitude masking; Temporal masking; Waveform coding; Perceptual coding; Codecs: MP3, AC-3, and Dolby Digital 2.3 Active Noise Control (ANC): Feedforward, feedback, and hybrid ANC; Secondary-Path Effects; LMS and FXLMS Algorithms; Deep ANC

	2.4 Audio Noise Reduction: Spectral subtraction; DNN-based approaches: Spectral masking; Denoising autoencoder; DCCRN; U-Net; Wavenet; Wave-U-Net								
	Laboratory Experiments: 1. Image processing and compression techniques 2. Digital Audio and Audio Enhancement								
Teaching/ Learning Methodology	Teaching and Learning Method	Intended Subject Learning Outcome	Remarks						
	Lectures	1, 2, 3	Fundamental principles and key concepts of the subject are delivered to students.						
	Tutorials	2, 3, 5	These are supplementary to lectures and are conducted with smaller class sizes; students will be able to clarify concepts and to gain a deeper understanding of the lecture material; problems and application examples are given and discussed.						
	Laboratory sessions	4, 5	Students will make use of software to simulate the various theories and visualize the results.						
Assessment Methods in Alignment with Intended Subject Learning Outcomes	Specific Assessment Methods/Tasks		% Weighting		Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)				
					1	2	3	4	5
	1. Continuous Assessment		50%						
	• Short quizzes		12%		✓	✓	✓		
	• Tests		24%		✓	✓	✓		
	• Laboratory sessions		14%		✓			✓	✓
	2. Examination		50%		✓	✓	✓	✓	✓
	Total		100 %						
	The continuous assessment will consist of a number of assignments, laboratory reports, and two tests.								
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:								

	Specific Assessment Methods/Tasks	Remark
	Short quizzes	These can measure the students' understanding of the theories and concepts as well as their comprehension of subject materials.
	Assignments, tests and examination	End-of chapter type problems are used to evaluate the students' ability in applying concepts and skills learnt in the classroom; students need to think critically and to learn independently in order to come up with an alternative solution to an existing problem.
	Laboratory sessions	Students are required to conduct some laboratory works, and produce the written reports; The accuracy and presentation of the report will be assessed; the emphasis is on assessing the students' ability to apply knowledge and skills learned in lectures, and their ability to relate the taken data and results to the most relevant theory.
Student Study Effort Expected	Class contact (time-tabled):	
	• Lecture	24 Hours
	• Tutorial/Laboratory/Practice Classes	15 Hours
	Other student study effort:	
	• Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes/examination	36 Hours
	• Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing	30 Hours
	Total student study effort:	105 Hours
Reading List and References	Textbooks:	
	1. R.C. Gonzalez and R.E. Woods, <i>Digital Image Processing</i>, 2nd ed., Prentice-Hall, 2002. 2. Ken C. Pohlmann, <i>Principles of Digital Audio</i>, 4th ed., McGraw-Hill, 2000.	
	Reference Books:	
	1. Ze-Nian Li and Mark S. Drew, <i>Fundamentals of Multimedia</i>, Pearson Prentice-Hall, 2004. 2. M. Mandal, <i>Multimedia Signals and Systems</i>, Kluwer Academic Publishers, 2003. 3. Sen M. Kuo and Dennis R. Morgan. "Active noise control: a tutorial review." <i>Proceedings of the IEEE</i> 87.6 (1999): 943-973. 4. Hao Zhang and DeLiang Wang. "Deep ANC: A deep learning approach to active noise control." <i>Neural Networks</i> 141 (2021): 1-10. 5. Loizou, Philipos C. <i>Speech enhancement: theory and practice</i>. CRC press, 2007.	